

Coding & Solution

Date	25 June 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	Website / github link
Programming Language(s)	Python 3.10+
Framework(s) Used	Flask, Scikit-learn
Key Features Implemented	• Crop recommendation based on 7 environmental parameters • K-Means Clustering with Elbow Method • Logistic Regression classification • Interactive web UI with prediction form • Model serialization with Pickle • Prediction history management
Pending / Incomplete Features	User authentication, cloud deployment, multi-language support
Setup / Run Instructions	1. Clone repo → 2. pip install -r requirements.txt → 3. python app.py → 4. Open http://127.0.0.1:5000 in browser

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

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